

Topic 10: Population Models, The Logistic Equation & Phase-Line Stability

Professor Leonard Master Series · Lessons 31 to 36

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Phase Lines & The Crowded Room

"What is an Autonomous Differential Equation? It is an equation where the rate of change depends ONLY on the current state y , not on time t : $\frac{dy}{dt} = f(y)$."

Think of a room with people:

- **Malthusian Exponential Growth:** $\frac{dP}{dt} = kP$. The more people, the faster it grows. But resources are not infinite!
- **Verhulst Logistic Growth:**

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right)$$

Here, K is the **Carrying Capacity**.

- When population is small ($P \ll K$), $(1 - P/K) \approx 1 \implies \frac{dP}{dt} \approx rP$ (exponential takeoff!).
- As population approaches K , $(1 - P/K) \rightarrow 0 \implies$ growth grinds to a halt!
- If population exceeds K ($P > K$), $(1 - P/K) < 0 \implies$ population starves and drops back towards K !

2. Curriculum Reference Diagram

$$\begin{aligned} \frac{d^2P}{dt^2} &= P \left(-b \frac{dP}{dt} \right) + (a - bP) \frac{dP}{dt} = \frac{dP}{dt} (a - 2bP) \\ &= P(a - bP)(a - 2bP) \\ &= 2b^2P \left(P - \frac{a}{b} \right) \left(P - \frac{a}{2b} \right). \end{aligned}$$

Figure 10.1: Logistic S-curve with inflection point at half the carrying capacity $P = K/2$ — from Textbook 7th Ed. Chapter 2 (Fig. 2.8.2).

3. Phase Lines & Stability Classification

Set $f(y) = 0$ to find the **Critical / Equilibrium Points**: Draw a vertical line (the phase line) marking the critical points:

1. **Asymptotically Stable (Attractor / Sink)**:

- Arrows point towards the critical point from both above and below.
- If perturbed, the system naturally returns to this state (like a ball at the bottom of a bowl).

2. **Unstable (Repeller / Source)**:

- Arrows point away from the critical point on both sides.
- Any tiny nudge causes the system to run away (like a pencil balanced on its tip).

3. **Semi-Stable**:

- Arrows point towards it from one side, and away from the other.